



6.6.6 Wrap-Up: Cool Down

1. Choose two of the visual models of subtraction that you are sure will result in a positive answer.

- a. Draw an asterisk (*) next to both in the table.

- b. Give an example of them below.

2. Choose one visual model that is a challenge for you.

- a. Draw an exclamation point (!) next to it in the table.

- b. Explain using an example why it is a challenge.

Visual Model of Subtraction

Unit 6, Lesson 7: Subtracting Integers Using Number Lines



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Targets

- ☐ I can subtract integers using a number line.
- ☐ I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.



6.7.1 Warm-Up: Bell Work

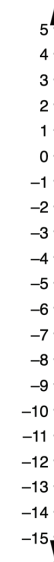
1. Draw integer chips that represent the expression below and evaluate.

$$7 - (-3)$$



6.7.2 Guided Instruction: Subtracting Integers Using a Number Line

1. The temperature in degrees Fahrenheit ($^{\circ}\text{F}$) of a city in Alaska was -4°F . After sunset, the temperature dropped 6°F .
 - a. Represent the change in temperature on the number line.
 - b. Write the subtraction number sentence that is represented from the context and on the number line.
 - c. After the decrease in temperature, what was the temperature?
 - d. Represent the temperature change using integer chips to verify the solution.



2. Complete the statements to explain how to represent subtraction on a horizontal number line.

a. To find the difference of $-9 - (-30)$, start by plotting

- ☐ 9
- ☐ -9
- ☐ 30
- ☐ -30

on the number line first.

b. The subtraction sign indicates moving to the

- ☐ right
- ☐ left

on the number line, but it is followed by

a negative sign with 30, which requires moving in the opposite direction.

c. Therefore, the answer will be found by moving

- ☐ 9
- ☐ 30

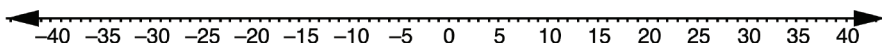
units to the

- ☐ right
- ☐ left

of -9 on the number

line.

3. Show the subtraction on the number line below.



The difference of $-9 - (-30)$ is _____.



6.7.3 Collaboration: Relating Integer Addition and Subtraction

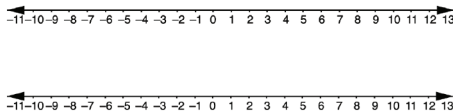
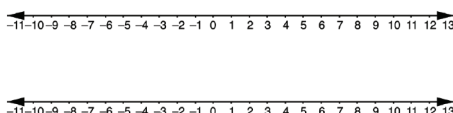
Three contexts are given below. Choose one context to work on with your partner. Circle your choice.

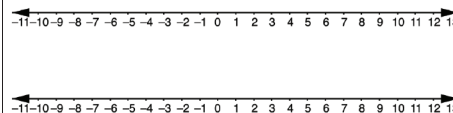
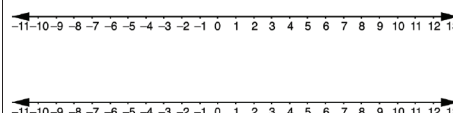
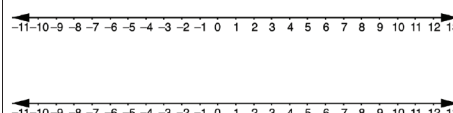
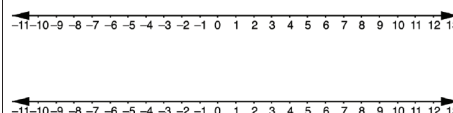
Context A	Tamara didn't realize she had a balance of \$19 in her bank account when she used her bank card to purchase a \$50 purse. When the purchase is processed by the bank, what will be her new account balance?
Context B	The elevation of the top floor of a parking garage in Death Valley, California, is 282 feet (ft) below sea level. A car that leaves the parking garage from the top floor loses 72 ft in elevation before reaching the street. What is the elevation of the street?
Context C	Ginny mows lawns for her summer job. Before she started mowing, she spent \$32 on gas for the lawn mower. Then, she spent another \$17 on other supplies she bought at the hardware store. What were Ginny's total expenses?

For the context selected, represent the mathematics in the context in each of the following ways.

1. Draw a number line to represent the given context.

- Write two equations that can be used to find the solution to the question in the context. One equation should use subtraction and the other should use addition.
- Create a group of four by joining with another pair of students. Verify your work by exchanging with the other pair of students.
- Continue working with the group of four to complete the table.

Example Subtraction Problem	Example Rewritten as an Addition Problem	Number Lines: Show the subtraction on the top number line and the addition on the bottom number line.
$10 - 7 = 3$	$10 + (-7) = 3$	
$7 - 10 = -3$		

$8 - -3 = 11$		
$-5 - 3 = -8$		
$-7 - -15 = 8$		
$-9 - -7 = -2$		



When subtracting integers, problems can be rewritten as an addition problem using the additive inverse of the second integer.

- Describe one advantage of changing a subtraction problem to an addition problem.
- Complete the table by rewriting the second integer in the subtraction model as its additive inverse and indicating the sign of the value of either expression. The first row has been completed. Once your group is done, split back into pairs and check your table with a different pair of students.

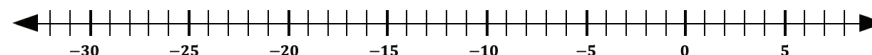
Visual Model of Subtraction	Visual Model of Addition	Sign of the Answer
		+



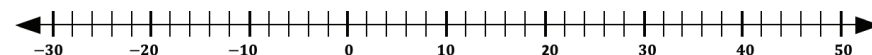
6.7.4 Your Turn!

- Represent each subtraction statement on a number line and find the value of the expression.

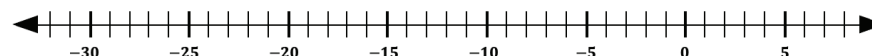
a. $-19 - 11 = \underline{\hspace{2cm}}$



b. $12 - (-28) = \underline{\hspace{2cm}}$



c. $(-14) - (-15) = \underline{\hspace{2cm}}$





6.7.5 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can subtract integers using a number line.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
$\longleftrightarrow$		

2. I can use number lines to understand how the rules for adding integers are related to the rules for subtracting integers.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.
$\longleftrightarrow$		

Unit 6, Lesson 8: Integer Subtraction



Benchmark

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.



Learning Target

- ☐ I can subtract integers with values greater than 100.



6.8.1 Warm-Up: Self-Reflection

For each statement, rate yourself on a scale of 1-5. A rating of 5 means “absolutely yes” and 1 means “absolutely not.”

Statement	Rating
I can subtract integers using integer chips.	
I understand how to show subtraction of integers on the number line.	
I can subtract integers without using integer chips or a number line.	
I can explain to someone else how to subtract integers.	



6.8.2 Guided Practice: Subtracting Integers

- Two expressions are shown.

Expression A	Expression B
$-191 + 77$	$-191 - (-77)$

With your partner, determine if the expressions are equivalent. Summarize your determination below.

- Annalise is on a trivia game show. Her score after the first round is -500 points. She misses the first question in the second round, losing an additional 125 points.

Kiersten said she could represent Annalise’s score using subtraction, while Dylan said it could only be represented with addition.

- Complete the table with the expression each student was thinking of.

Kiersten’s Subtraction Expression	Dylan’s Addition Expression
_____ - _____	_____ + _____

- Kiersten said Annalise’s score was -375 after missing the first question in the second round, but Dylan said it was -625 . Explain which student is correct.

3. Rewrite each subtraction equation as an equivalent addition equation.

Subtraction Equation	$513 - 382 = 131$	$957 - ^{-}136 = 1,093$
Addition Equation		

Addition and subtraction are **inverse operations**, which means they are operations that have the opposite effect of each other.

Understanding this relationship and the existence of additive inverses can be helpful when subtracting integers.



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

4. Complete the statements below using this information.

A subtraction expression can be rewritten as an equivalent addition expression by changing the operation from subtraction to _____ and changing the second value to its _____.



6.8.3 Try It: Subtracting Integers

1. The table shows the beginning and ending elevations, in feet (ft), of six different hiking trails. The change in elevation along the trail can be found by subtracting the initial elevation from the final elevation.

$$(\text{final elevation}) - (\text{initial elevation}) = (\text{change in elevation})$$

- a. Write an expression that represents the change in elevation along each hiking trail. Then, write the value that represents the change in elevation.

The first row has been completed as an example.

Initial Elevation (ft)	Final Elevation (ft)	Difference Between Final and Beginning	Change in Elevation (ft)
400	900	$900 - 400$	500
115	68		
294	-121		
-200	610		
-163	-57		
-17	-87		

- b. Explain what a negative change in elevation means in this context.



2. Find each difference.

a. $-664 - 231$

b. $109 - (-115)$



6.8.4 Your Turn!

1. Evaluate each expression.

a. $740 - (-171)$

b. $569 - 143$

c. $-738 - (-212)$

d. $333 - (-204)$

e. $-877 - 814$

f. $-691 - 293$

g. $157 - (-120)$

h. $341 - (-504)$



6.8.5 Wrap-Up: Ticket Out the Door

1. Evaluate each expression.

a. $447 - (-702)$

b. $-834 - (-610)$

c. $-309 - 525$

Unit 6, Lesson 9: Multiplying Integers Using Models



Benchmark

MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.



Learning Target

- ☐ I can multiply integers using integer chips, number lines, and patterns.